

FinVerify Optimization

SQL Analysis Task 2 of 3

KYC Failure Deep-Dive

Breaks down KYC failure reasons by document type in April 2026, excluding users who eventually passed on a later attempt.

SECTION 01: SQL Query

```
WITH failed_attempts AS ( SELECT user_id, document_type, vendor_status FROM kyc_attempts WHERE
verification_timestamp >= '2026-04-01' AND verification_timestamp < '2026-05-01' AND
vendor_status IN ( 'failed_clear_error', 'failed_suspected_fraud' ) AND user_id NOT IN (
SELECT user_id FROM kyc_attempts WHERE vendor_status = 'passed' ) ) SELECT document_type,
vendor_status AS failure_type, COUNT(*) AS failure_count, ROUND( COUNT(*) * 100.0 /
SUM(COUNT(*)) OVER (PARTITION BY document_type), 2 ) AS failure_percentage FROM
failed_attempts GROUP BY document_type, vendor_status ORDER BY document_type,
failure_percentage DESC;
```

SECTION 02: Expected Output

Actual output values depend on the underlying dataset. The table below illustrates the expected output format.

document_type	failure_type	failure_count	failure_percentage
Passport	failed_clear_error	120	75.00
Passport	failed_suspected_fraud	40	25.00
Driver's License	failed_clear_error	90	60.00
Driver's License	failed_suspected_fraud	60	40.00

SECTION 03: Query Explanation

Step 1 — CTE (WITH failed_attempts AS ...): We isolate only the failed KYC records from April 2026. The vendor_status IN (...) filter picks only the two failure codes: failed_clear_error (image too blurry or document unreadable) and failed_suspected_fraud (document suspected to be fake).

Step 2 — Excluding Eventual Passers: The AND user_id NOT IN (SELECT user_id ... WHERE vendor_status = 'passed') sub-query removes any user who eventually got a 'passed' result on a later attempt — regardless of when. This ensures we only analyse permanently failing users, not temporary failures.

Step 3 — Counting and Grouping: The outer query groups by document_type and vendor_status and counts total failure records per group using COUNT(*).

Step 4 — Window Function for Percentage: SUM(COUNT(*)) OVER (PARTITION BY document_type) calculates the total failures for each document type across all failure types. Dividing each group's count by this

total gives the percentage split within each document type, enabling apples-to-apples comparisons.

SECTION 04: Business Impact

01 Target Retry Campaigns: Knowing that 75% of Passport failures are clear errors lets the team send tailored 'please retake your photo' nudges instead of generic messages.

02 Flag Fraud Concentrations: If Driver's Licence shows a disproportionately high suspected-fraud rate, compliance teams can tighten manual review rules for that document type.

03 Vendor Accountability: Sharing this breakdown with FinVerify gives them concrete data to improve their OCR accuracy for specific document formats.

04 Product Roadmap Prioritisation: High clear-error rates justify investing in better in-app camera guidance (e.g. real-time image quality feedback before submission).